

Improved surrogate modeling for multi-energy system design: Model architecture, sampling and scaling choices

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HIGHLIGHTS

- Energy system design shows irregularities and sudden changes in design variables.
- Irregularities and splits reduce the learnability via neural-network.
- A large exploration of modeling parameters is tested over a large pool of cases.
- A simple mixture-of-experts architecture often addresses the issue.
- Upsampling and a novel sampling strategy support the surrogate model's training.

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ABSTRACT

Multi-energy systems (MES) are a key concept for developing more sustainable energy systems, but optimizing their design is computationally burdensome. This paper explores the development of machine-learning (ML) based surrogate models for the optimal design of MES. Surrogates are simple models, often ML-based, used to approximate detailed simulations, in this case MES design optimizations. These models provide instant responses, enabling fast comparisons and explorations of trade-offs between design variables. No related work proposes an ML procedure tailored to properties of the MES design application. Most related works use surrogates to predict system cost and other objectives. However, few works have used them to directly predict the optimal system design, and those that do show poor performance.

This paper provides an extensive methodology tailored to properties of MES design problems to improve surrogate performance on small datasets. Four components were found to significantly improve surrogate performance: a careful and objective-oriented selection of samples, the use of upsampling to balance datasets, the use of non-linear rescaling methods, and a specific neural-network architecture called Mixture-of-Experts. These work together to turn the original design variable distribution (i.e., of the output) into a Gaussian-like data distribution, that can be more easily learned by the neural-network.

The resulting surrogate model almost instantly predicts optimal energy system designs with high precision. This was tested across a wide variety of different climates, building types and decarbonization goals. Such surrogate models will make it much easier to explore different MES design options.

1. Introduction

The transition towards sustainable energy systems requires innovative approaches to optimize the integration of diverse resources and build robust multi-energy systems (MES) [1]. Advanced modeling techniques, for example the “Energy Hub” (EHub) concept [2], were found

to be relevant from early design stages to identify synergies between diverse technologies and energy sources to be exploited.

Traditionally, the optimization of MES models involves formulating complex mathematical models representing the interactions among different components and solving them using conventional optimization techniques [1]. However, the computational burden associated with

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solving these models grows exponentially with the system size [3] and level of detail [4,5], thus limiting the possibility of scaling models up.

In recent years, machine learning (ML) techniques and the concept of surrogate modeling have been explored to address challenges related to the design [6] and operation [7] of MES. Surrogate models aim at replacing a detailed and computationally burdensome task with a fast estimate, often ML based [8,9]. Applications of interest lie in early stage design exploration and simulation intensive sensitivity analyzes [8,10], thus the acceptability of the time vs. accuracy trade-off. For instance, surrogates of building energy simulations are common [11–14].

The most common usage of surrogate models for MES design is called surrogate-assisted modeling. It is a bi-level optimization approach [15]: on the first level, an evolutionary algorithm (e.g., GA, PSO) suggests MES designs (i.e., selection and sizing of generators and storages); on the second level, a surrogate evaluates the potential performance of the design by estimating the costs [16–18], emissions [16,19], and/or other objectives [20–22] related to the system operation.

Surrogates fundamentally specialize in linking a set of model parameters with a set of model responses [23], but do not offer guarantees of performance on subregions of interest [14,24]. Where the indirect design suggested in [22,25] requires surrogates performing well for a diversity of designs (including absurd designs), nothing guarantees the required accuracy near the optimum or close to non-dominated solutions.

To the best of the author's knowledge, only four works have attempted to surrogate the whole MES design process. Köhnen et al. [26] compare neural-network (NN) architectures to estimate capacities and total energy supply for an MES. Jahangiri et al. [27] build surrogates with a deep NN for a grid-expansion planning model of Canada, then conduct a sensitivity analysis with the surrogate. Thrampoulidis et al. [28] build surrogates with NNs for an MES model addressing capacity sizing and selection of retrofit packages for residential buildings in Zurich. To define a cost-carbon Pareto front (PF) of optimal solutions, the NNs are chained and use both data and predictions from the previous NN to estimate the optimal design at the next point of the PF. The authors extend the surrogate capabilities to the whole of Switzerland in [29], by replacing each NN with a super-structure of six intertwined NNs. Each of the two regressors and four classifiers addresses a specific set of variables and benefits from the predictions of other NNs within the super-structure.

A common limitation found across [26,28,29] is the large difference in performance between the predictions of some variables. While surrogates can predict the costs and emissions (i.e., The objective values) and some capacities with great precision ($R^2 \geq 0.95$), they systematically fail ($R^2 \leq 0.7$) at predicting other capacities (e.g., photovoltaics, heat pump and storage sizes). Results from [26] also reveal that the random weight-initialization before the training of NN may lead to significantly different performance between two repetitions of the same task, especially on the variables yielding insufficient performances. These limitations prevent the full deployment of a functional surrogate, but are neither addressed nor discussed.

This paper investigates the possible reasons for these discrepancies, and characterizes a set of methodological solutions to build more reliable surrogates of MES for design applications. Accurately predicting the optimal capacities of the components of an MES is the primary purpose of a design optimization; predicting just the objectives with no knowledge of the capacities, or predicting capacities with high errors, is not sufficient. The novel methodology developed in this paper particularly focuses on predicting all component capacities. To ensure its robustness and generalizability, the methodology has been tested across diverse climates, building types and problem constraints (e.g., desired carbon emissions limits).

The document is structured as follows. Section 2 introduces the case study (core application, model, variables of interest and data used) and presents the core hypothesis driving the study. The new process

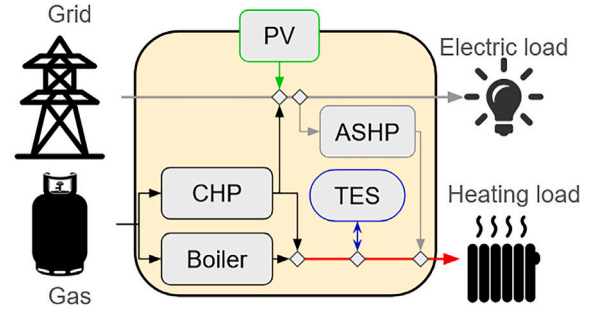


Fig. 1. Potential layout of the studied energy system.

suggested to train surrogate models and address the hypothesis is presented in great detail in Section 3. Results are presented and discussed in Section 4, investigating whether the new implementation addresses the issue, how it addresses the issue and why it is successful.

2. Multi-energy system design problem formulation

The methodology and results presented in this paper address the MES design problem for the system presented in Fig. 1. The model formulation, variables of interest and data used are respectively described in Sections 2.1, 2.2 and 2.3. A specific property of the application is highlighted in Section 2.4, where the core hypothesis behind this research is formulated.

2.1. Energy system model

The core MES is designed following the concept of the Energy Hub [2]. The design application is formulated as a mixed-integer linear programming (MILP) problem, as largely covered in related literature [1]. A mathematical optimizer optimally selects and sizes the assets presented in Fig. 1, and suggests an optimal hourly control strategy to operate the assets (purchase electricity and gas, store, reconstitute and transform energy) to cover the hourly heating and electric load of some building for a whole year (8760 time steps). Gurobi 12 software [30] is used for this task. The suggested sizing and operation strategy aims to minimize the annualized cost under some carbon emission reduction target. The few following equations are the main equations leading to the outputs of interest for this study.

The system design assumes an underlying hourly operation, during which energy can be bought ($\dot{Q}_{\sigma,t}^{N,in}$), sold ($\dot{Q}_{\sigma,t}^{N,out}$), transformed ($\dot{Q}_{\sigma,t}^{T,in}$), stored ($\dot{Q}_{\sigma,t}^{ch}$) and consumed ($\dot{Q}_{\sigma,t}^{load}$) on different energy streams ($\sigma \in S$) at every time step ($t \in \mathbb{T}$), following (Eq. 1).

$$\begin{aligned} \dot{Q}_{\sigma,t}^{load} = & \sum_{\tau_{cv} \in \mathcal{T}_{cv}} \left(H_{\tau_{cv},\sigma} \dot{Q}_{\tau_{cv},t}^{T,in} \right) + \dot{Q}_{\sigma,t}^{N,in} - \dot{Q}_{\sigma,t}^{N,out} \\ & + \sum_{\tau_s \in \mathcal{T}_s,\sigma} \left(\eta_{\tau_s}^{dch} \dot{Q}_{\sigma,\tau_s,t}^{dch} - \eta_{\tau_s}^{ch} \dot{Q}_{\sigma,\tau_s,t}^{ch} \right) \end{aligned} \quad (1)$$

Two outputs of interest are the annualized cost (Eq. 2) and yearly carbon emissions (Eq. 3) across four scenarios enforcing different carbon emission limits (Eq. 4) [31]. Both are related to the system operation, particularly the energy purchases ($\dot{Q}_{\sigma,t}^{N,in}$). The cost also involves the net-present value (NPV) for the annualization, fixed prices (C_{τ}^{fix}), linear prices (C_{τ}^{lin}) and capacities of devices (P_{τ}^{nom}), as well as prices for maintenance (C_{τ}^{mnt}) and carbon emissions (C^{CO_2}). The calculation of emissions involve the carbon factors G_f of the considered sources $f \in \mathcal{F}$.

$$J_{tot}^a = \left(\begin{array}{l} \sum_{f \in \mathcal{F}} C_f^{fuel} (\dot{Q}_{\sigma_f}^{N,in} - \dot{Q}_{\sigma_f}^{N,out}) \\ + \sum_{\tau \in \mathcal{T}} NP V_{\tau} (C_{\tau}^{fix} + P_{\tau}^{nom} C_{\tau}^{lin}) \\ + C_{CO_2} J_{CO_2} \\ + \sum_{\tau \in \mathcal{T}} C_{\tau}^{mnt} P_{\tau}^{nom} \end{array} \right) \quad (2)$$

$$J_{CO_2} = \sum_{f \in \mathcal{F}} G_f (\dot{Q}_{\sigma_f}^{N,in} - \dot{Q}_{\sigma_f}^{N,out}) \quad (3)$$

The generation of solutions along the carbon limitation scenarios is done in two steps: (1) both solutions for minimum cost and minimum emissions are computed, respectively, by using the annualized cost (Eq. 2) and carbon emissions (Eq. 3) as objective functions of 2 different problems. The carbon emissions of both solutions are kept as bounds. (2) Then the points constrained to different carbon emission limitations are computed by minimizing the annualized cost while enforcing various upper limits X_{CO_2} on the carbon emissions (Eq. 4). The value of these limits are set between the two emission bounds from step 1 (c.f. list in Section 2.3).

$$J_{CO_2} \leq X_{CO_2} \quad (4)$$

The capacity of a converter $\tau_{cv} \in \mathcal{T}_{cv}$ is defined by its maximum generation during hourly operation of the whole system (Eq. 5), conditioned by the conversion efficiency $H_{(\tau_{cv}, \sigma_{ref})}$. Non-dispatchable converters like PV constantly operate at full capacity given the resource availabilities $\xi_{PV,t}^{sun}$. Regarding the storage $\tau_s \in \mathcal{T}_s$, the state of charge (SOC) cannot exceed the device capacity (Eq. 6), and the charged/discharged energy within an hour is limited by a fraction $\beta_{\tau_s}^{ch/dch}$ of the storage capacity (Eq. 7). The state of charge is propagated throughout time steps following Eq. (8) and forced to describe a yearly cycle (Eq. 9).

$$\begin{cases} 0 \leq H_{(\tau_{cv}, \sigma_{ref})} \dot{Q}_{\tau_{cv},t}^{T,in} \leq P_{n,\tau_{cv},t}^{nom}, \\ \quad \text{if } \tau_{cv} \text{ is dispatchable} \\ H_{(PV, \sigma_{ref})} \dot{Q}_{PV,t}^{T,in} = \xi_{PV,t}^{sun} P_{PV}^{nom}, \end{cases} \quad (5)$$

$$0 \leq SOC_{\sigma, \tau_s, t} \leq P_{\tau_s}^{nom} \quad (6)$$

$$\left\{ 0 \leq \dot{Q}_{\sigma, \tau_s, t}^{ch} \leq \beta_{\tau_s}^{ch} \times P_{\tau_s}^{nom} \right. \quad \left. 0 \leq \dot{Q}_{\sigma, \tau_s, t}^{dch} \leq \beta_{\tau_s}^{dch} \times P_{\tau_s}^{nom} \right. \quad (7)$$

$$SOC_{\sigma, \tau_s, t+1} = \eta_{\tau_s} SOC_{\sigma, \tau_s, t} + \eta_{\tau_s}^{ch} \dot{Q}_{\sigma, \tau_s, t}^{ch} - \frac{1}{\eta_{\tau_s}^{dch}} \dot{Q}_{\sigma, \tau_s, t}^{dch} \quad (8)$$

$$SOC_{\sigma, \tau_s, t^{max}+1} = SOC_{\sigma, \tau_s, t=0} \quad (9)$$

2.2. Variables of interest and ranges

The system is kept simple for the following reasons; (a) it is generic enough to be suited for diverse building types and locations at early design stage; (b) it facilitates the exploration and understanding of design decisions; (c) it limits the number of design variables of interest, thus more focus can be put towards the surrogates. Despite its simplicity, such a system presents many design choices that may be optimal in some contexts or suboptimal in others. The surrogate modeling task is challenging, even for such a simple case. The choice between technologies may result in splits in the distribution of design variables, which is believed to limit the performance of the surrogate models.

Table 1
Surrogate model output space.

Variable	Name	Unit	Additional description
Total cost (J_{tot}^a)	Cost	k\$/y	Sum of expenses (Eq. 2)
Emissions (J_{CO_2})	Carbon	tCO ₂ eq/y	Sum of emissions (Eq. 3)
ASHP capacity (P_{ASHP}^{nom})	ASHP	kW _{th}	-
Boiler capacity (P_{Boiler}^{nom})	Boiler	kW _{th}	-
CHP capacity (P_{CHP}^{nom})	CHP	kW _{el}	-
TES capacity (P_{TES}^{nom})	TES	kWh	-
TES usage ($\sum_{t \in \mathcal{T}} \dot{Q}_{TES,t}^{dch}$)	TesUse	MWh/y	Yearly TES discharge
Electricity ($\sum_{t \in \mathcal{T}} \dot{Q}_{Elec,t}^{N,in}$)	Elec	MWh/y	One year of grid import
Gas ($\sum_{t \in \mathcal{T}} \dot{Q}_{Gas,t}^{N,in}$)	Gas	MWh/y	One year of gas import

Table 2
Ranges and bounds of the considered input space.

Variable	Range
Price of grid electricity C_{elec}^{fuel}	[0.15, 0.35] \$/kWh [33]
Price of gas C_{gas}^{fuel}	[0.04, 0.08] \$/kWh [34]
Carbon tax C_{CO_2}	[50, 200] \$/tCO ₂ eq [27]
Carbon intensity of grid electricity G_{elec}	[150, 700] gCO ₂ eq/kWh [35]
Carbon intensity of gas G_{gas}	[219, 288] gCO ₂ eq/kWh [36,37]

Table 3
Fixed parameters of the MES design problem.

Device	Life (y)	C^{fix}	C^{lin}	H/η	$\eta^{ch/dch}$	$\beta^{ch/dch}$
ASHP	20	2968	198	pre	-	-
Boiler	25	1492	47	97 %	-	-
CHP	25	1283	246	35 %/45 %	-	-
PV	15	0	1020	pre	-	-
TES	25	400	99	98 %	100 %	40 %

The nine design variables of interest are introduced in Table 1. The cost and carbon are commonly used as objective values to define Pareto Fronts in the related literature, and are of primal interest to evaluate the overall system. The capacities of diverse technologies are often the main usage of EHub and indicate how to achieve least cost and emissions. The total energy purchases reveal the degree of electrification of the system, and are a main component of the design process and of the cost calculations. The total usage of storage informs about the overall need to shift the demand for better performance.

The core MES design problem has many parameters describing the context (e.g., costs of technology, price of fuels, conversion efficiencies). Five context variables, found to be most sensitive over reasonable ranges by previous research [32], are selected to define the input space: the carbon tax, the prices of electricity and gas, the carbon intensities of electricity and gas. The selected ranges of values are typical for a North American context (Table 2). One unique value is considered throughout the whole year for the sake of simplicity. Values for other model parameters not included in the input space are presented in Table 3, where *pre* indicates a pre-calculation with hourly resolution.

For the carbon intensity of gas, Venkatesh et al. [36] and Burnham et al. [37] provide LCA-based estimates, ranging at 219–260 gCO₂eq/kWh for the former, and either 252–288 gCO₂eq/kWh or 324–414 gCO₂eq/kWh depending on the time horizon considered (100 years and 20 years) for the latter.

2.3. Data and context

The energy loads used for this research are taken from a dataset of building energy simulations for different building types across the US [38]. A total of nine cases are selected out of the original dataset, exhaustively combining the locations (from different climate zones) and building types in the list below. For each case, the system in Fig. 1 is designed optimally according to Eq. (4), with each relative decarbonization

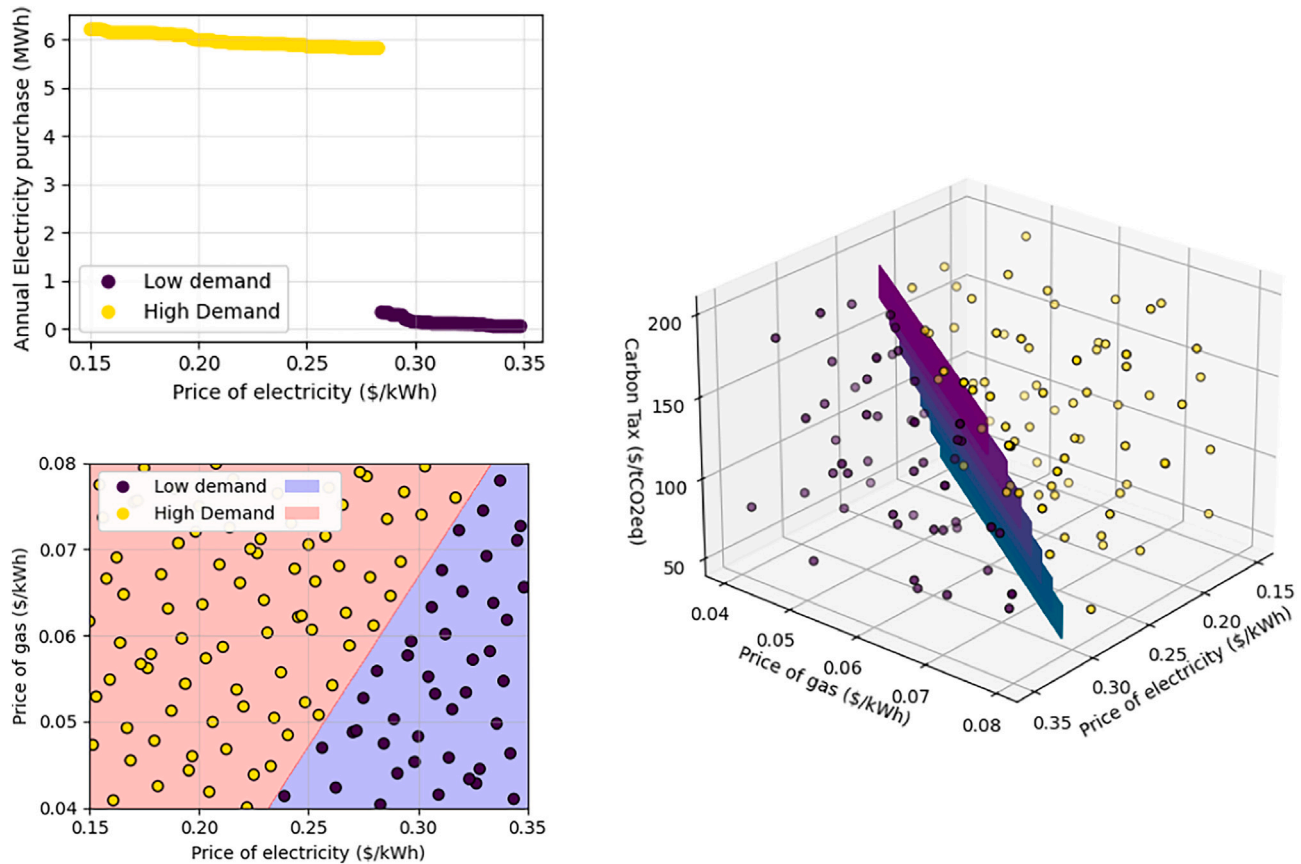


Fig. 2. Illustration of splits occurring in the design variable on simplified problems with one (top left), two (bottom left) and three (right) variables.

objective presented in the list below. With 9 contexts, 4 decarbonization objectives and 9 outputs of interest, a total of 324 independent datasets/experiments are considered.

- **Locations:** Helena, MT; San Francisco, CA; Miami, FL.
- **Building type:** Residential (SFH); Hospital; Office building.
- **Relative decarbonization goal:** 0 % (minimum cost); −50 %; −90 %; −100 % (minimum CO₂).

The diversity of contexts results in a pool of datasets exhibiting a large variety of properties over the considered ranges (Table 2) and covered climate zones. These properties are of prime interest in the remainder of the study.

2.4. Motivational example

The base model selects technologies and sizes them to match energy demands. In many cases, and independently of the context, a design variable of interest may exhibit a split distribution due to that selection. Fig. 2 illustrates such a phenomenon in the design of an ASHP, through simplified problems using only one, two and three input variables. As shown in Fig. 2, the split is non-trivial, i.e., it depends on all variables involved in the problem.

The methodology presented later in the current research is based on the hypothesis that ML-based surrogates learn better when the target values exhibit a Gaussian-like distribution, or at least a continuous and balanced distribution. Therefore, the suggested methodology focuses on ways to make the data perceived by the surrogate as balanced and continuous as possible. Mainly, the focus is directed towards addressing potential splits in the data, with a possible imbalance in the number of samples on either side of a split.

3. Methodology

Fig. 3 highlights the main steps in the process of building the surrogate models. The general pipeline is straight-forward: data is generated, a specific model type is selected, tuned via hyperparameter optimization (HPO), the final model is re-trained and evaluated. This section introduces a novel sampling step in the data generation (Section 3.1), the different model types used (Section 3.2), the range of parameter options during the HPO (Section 3.3) and the evaluation procedure (Section 3.4).

3.1. Data generation

The selection of samples is based on a Halton sequence [39]. Each set of inputs is used to design the energy system via the EHub application. The resulting dataset is split evenly into a training set, a test set and a validation set, all three with 256 samples each. The training set may then be subjected to additional upsampling. The choices of sampling sequence and of even size of training–testing–validation sets are discussed in Section 3.1.1. Section 3.1.2 explains the sampling approach. Section 3.1.3 introduces the upsampling step.

3.1.1. Training, test and validation sets

All samples are drawn with a Halton sequence [39], a low-discrepancy sequence. Preparatory work revealed that the discrepancy of subsequences is more homogeneous than that of the Sobol sequence [40]. Therefore, training, test and validation sets of similar size and issued from subsequences of the same series guarantee a similar coverage, with similar properties but no repetition.

Similar properties are needed for better training and evaluation. Splits and imbalances are expected in the distributions of design variables (c.f. Fig. 2), however their occurrence or location is unknown prior

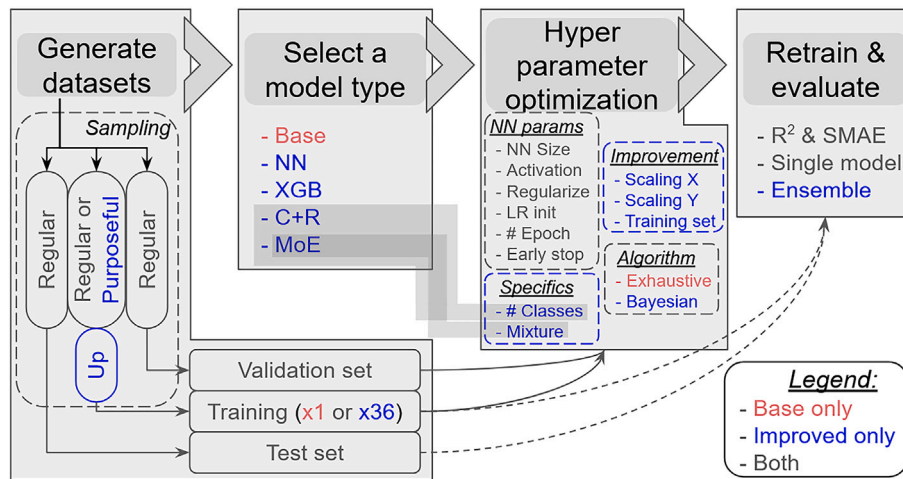


Fig. 3. Overview of the procedure to train the surrogate models.

to computation of the original application. If the density of samples near a split is insufficient:

- ... in the test set, the performance evaluation will not correctly incorporate the difficulty of the design space, despite modeling efforts.
- ... in the validation set, a suboptimal selection of hyperparameters is likely, and the resulting model may perform poorly on unseen samples near the split.

3.1.2. Sampling

A pool of 16 blocks is drawn from a Halton sequence, each block containing 128 points. Blocks #15 and #16 are used as test set, and blocks #13 and #14 as validation set. The other blocks may be used for the training set via one of the two following approaches.

The first approach merges blocks #1 and #2 to form the training set. It is the standard approach, used for example in all base cases.

The second approach, illustrated in Fig. 4, is named “purposeful sampling” in this paper, and is only available in the improved approach. It is inspired by the concept of “border patterns” [41] to adapt the purposeful sampling method common in qualitative surveying [42] to the current regression problem. Labels from block #1 are grouped into clusters (using Gaussian Mixture clustering). An artificial neural network (ANN) is then trained to classify samples from block #1 into their assigned cluster. The classifier is called “gating” due to similarities with a model introduced in Section 3.2. The classifier is then used on blocks #2 to #12 to identify the 128 points least likely to be correctly classified (X_+). Distinctions between 2, 3 and 4 clusters are explored in this paper.

3.1.3. Up-sampling

The training set may be subject to upsampling, in the improved approach only. Upsampling adds synthetic information (i.e., not processed through the original simulation) in specific subregions of the dataset. This is commonly used to enhance the number of samples in underpopulated classes for classification. The rare usages for regression flag “exceptional” points, then interpolate between them with various methods.

The hyperparameter optimization chooses whether to perform upsampling, then selects one among eight interpolation methods. These methods are the available options from the Python package ImbalancedLearningRegression [43].

3.2. Surrogate model types

This work uses two model types: ANNs on which all classifiers and most regressors rely, and Tree Boosting models as regressors for the sake

of comparison. The base neural network (NN) and fine NN are both ANNs (Section 3.2.1). A Classifier–Regressor enhances the input space of a regressor with predictions from a classifier (Section 3.2.2). A Mixture of Experts chains a classifier with multiple independent regressors (Section 3.2.3). Tree Boosting uses an ensemble of decision trees (Section 3.2.4). The training and predicted data are rescaled independently for each ANN composing the different models.

The ANNs, ANN-based models and clusterings use the `scikit-learn`¹ package. This work also uses a gradient boosting decision tree (GBDT) model from `scikit-learn` and an extreme gradient boosting (XGB) model from `XGBoost`² [44].

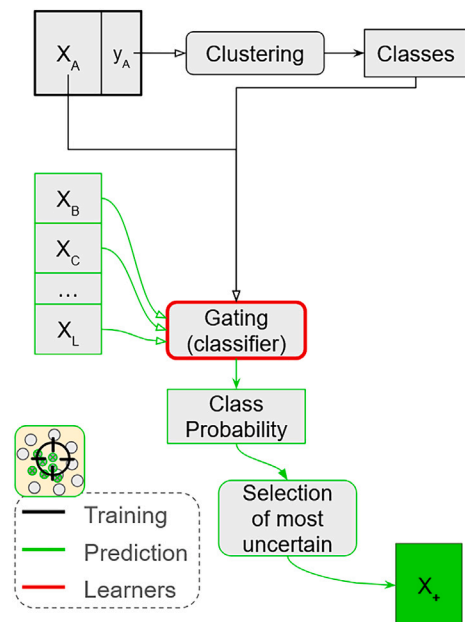


Fig. 4. Methodology for the purposeful sampling.

¹ <https://scikit-learn.org>.

² <https://xgboost.readthedocs.io/en/stable/>.

3.2.1. Base and improved NN models

The experiment suggests a base and an improved NN model, each made of one single base ANN regressor. The former is trained via the base methodology, the latter via the improved methodology. Both choose among similar architectures (Table 4), but they differ in the number of options for the usable training sets, data preprocessing, a few discrete hyperparameters, and the algorithm conducting the hyperparameter optimization (Section 3.3).

Table 4
Architecture and corresponding number of neurons in the hidden layers (hl).

Architecture	Number of neurons		
	hl1	hl2	hl3
Small	16		
Shallow	128		
Medium	6	36	6
Large	16	64	16

3.2.2. Classification–regression

A classification–regression (C+R) model [29,45,46] chains together a classifier and a regressor, so that the prediction from the classifier is used as additional input information by the regressor (Fig. 5). First, Gaussian Mixture clustering distinguishes 2, 3 or 4 groups within the training outputs y . A classifier is then trained to predict the class labels from the training inputs X . These inputs X and class labels are also merged to train the regressor to predict the training outputs y . In prediction mode, the class probabilities $\hat{y}$ predicted by the classifier are merged with the input data to inform the regressor, yielding the final prediction $\hat{y}$.

3.2.3. Mixture of experts

A Mixture-of-Experts (MoE) is an ensemble model following a divide-and-conquer approach [47]. It subdivides the input space with a gating unit and delivers more localized predictions with expert units. Such architecture can address the possible abrupt jumps [48] observed in the

design variables. The gating, experts, their interconnection and training take different forms in the literature. Any probabilistic model or supervised ML algorithm is suitable as gating and expert [47,49]; their training may be separated [49,50] or addressed altogether [47], clustering may support the subdivision process [49,51], and the prediction may be handled as a mixture of all experts' opinions [49,50,52] or by the most trusted one in each subsection [53].

The proposed MoE implementation (Fig. 6) clusters the training outputs y into 2, 3 or 4 groups using Gaussian Mixture [51]. A classifier (i.e., The gating) is trained to infer cluster labels from the training inputs X . The generated classes are also used to segment the training set into subsets, each subset being used to train independently a different expert regressor.

In prediction mode, the set of new inputs X_{new} is used through each expert to generate every expert's "opinion" $\hat{y}^k$, and through the gating to generate a level of trust in each expert (probability $\hat{y}^k$ of a sample to belong to each class). As additional hyperparameter, the prediction for each sample may either consider only the most trusted expert, or consider an average of all experts' opinions, weighted by the relative probabilities.

3.2.4. Tree boosting

Boosting aims to create "strong" learners from an ensemble of base learners [54]. Base learners are simple, non-robust models. The most common is the Decision Tree, e.g., for Gradient Boosting Decision Trees (GBDT) [55] or eXtreme Gradient Boosting (XGB) [44]. These methods give state-of-the-art results in many machine-learning competitions (e.g., The ASHRAE Great Energy Predictor III Competition [56]). Moreover, decision trees essentially work by identifying splits in the data. Therefore, they appear as promising candidates for the expected difficulties exposed in Section 2.4.

A first base learner is trained. Boosting then iteratively adds and sums up new base learners to reduce the errors from the previous ensemble [53]. As base learners are subject to overfitting [57], various regularization techniques are used. A decreasing weight factor (learning rate) is assigned to each learner. Learners are presented with only a subset of the training data, and a subset of the available features. Base tree learners may be limited in depth, in number of nodes, or by constraints to allow

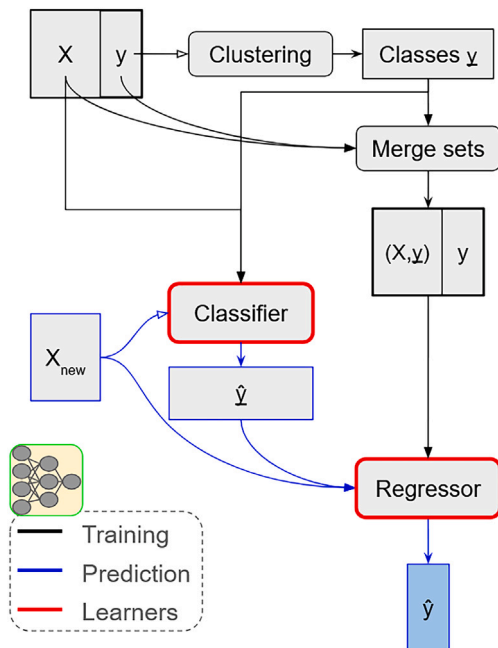


Fig. 5. Methodology for the classification–regression.

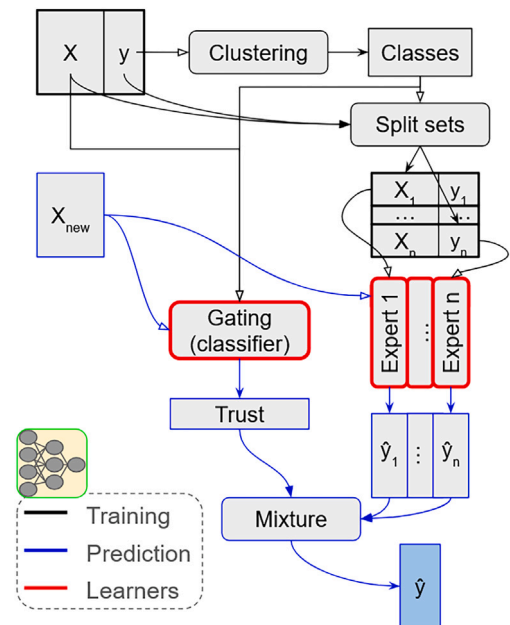


Fig. 6. Methodology for the mixture-of-experts.

creating branches. Both a GBDT and an XGB are used in this work, as the training, regularization approaches, and set of hyperparameters differ.

3.2.5. Ensembles

Known as a robust solution in machine learning problems, training an ensemble of neural networks and averaging the predictions perform better on numerous machine learning problems [58,59], particularly when dealing with small datasets.

Every model is trained and evaluated four times as part of the evaluation procedure, a last step of the procedure thus evaluates the averaged prediction of all four models. For regressors, the prediction for each sample is averaged (Eq. 10). For classifiers, the predicted probability of each class is averaged following Eq. (10) before the class estimated to be most likely is considered the final class prediction.

$$\hat{y}_{\Omega} = \frac{1}{4} \sum_{i=1}^4 \hat{y}_i \quad (10)$$

3.2.6. Data scaling

Scaling data helps training ANNs effectively. Standardization, i.e., subtracting the mean and dividing by the standard deviation, is commonly used [60]. Logarithmic [61] or power transformations (e.g., box-cox [13,62]) may improve performance with skewed or imbalanced data, by reshaping it into a more Gaussian-like distribution. Quantile transformation [63] does the same by working on quantiles. This non-parametric approach is particularly suited to surrogate modeling, where ranges of values are fixed. It is therefore preferred to the power transformations in this study.

In this study, the improved methodology compares standardization with quantile transformations creating normal and uniform distributions. Inputs (X) and outputs (y) of each base learner can be scaled differently and independently. For example, an MoE can apply distinct scaling for the gating function and for each expert. Only standardization is used with the base methodology.

3.3. Hyperparameter optimization

Hyperparameter optimization (HPO) is a process that aims at selecting the combination of hyperparameters to maximize the model performance. Different methodologies exist to perform this task. An exhaustive search is used for the base model (i.e., all combinations of available options for each hyperparameter are compared). Bayesian optimization [64] is used for the improved models. For the non-common hyperparameters (e.g., architecture, scaling, sampling, upsampling) the HPO algorithm selects the value of an intermediate categorical variable, this value is later interpreted to refer to the correct action in the problem build up. The implementation from `scikit-optimize`³ is used for the Bayesian HPO, and those from `scikit-learn`⁴ for the exhaustive search.

Tables 5 and 6 lists all sets and ranges of options to select from during the HPO.⁵ The check mark ✓ indicates that the option, set or range is considered entirely. A value indicates that other options are discarded. A ~~struck~~ value indicates that all but this option are considered. Specifically for MoE, the asterisk * indicates that the option, set or range is considered for each expert independently. The double-asterisk ** indicates that separate models were trained and evaluated for each option. Similar classifiers (Clf) are used for the C+R and for the gating of MoE. They undergo a HPO separately from the regression part, but are presented together in Table 5.

Hyperparameters and ranges used for the tree boosting models in Table 6 are based on a related blog [65] and literature [66–69].

Table 5

Hyperparameters and ranges for the ANN-based models.

Name	Options/Range	Base	NN	C + R		MoE
				Reg.	Clf.	Reg.
Neural network hyperparameters						
Architecture	Small				✓	*
	Shallow	✓	✓	✓	✓	*
	Medium	✓	✓	✓	✓	*
	Large	✓	✓	✓		
Activation	{Linear, ReLu, TanH}	✓	✓	✓	✓	*
L2 regularization	$\log(\alpha) \in [-5, -1]$	✓	✓	✓	✓	*
Learning rate init	$\log(lr0) \in [-3, -1]$	✓	✓	✓	✓	*
Epochs	{200, 1500, 5000}	1500	✓	✓	✓	5000*
Early stopping	{8, 16, 32, 64}	32	8-	8-	8-	64*
Tolerance	10^{-5}	✓	✓	✓	✓	✓
Solver	Adam	✓	✓	✓	✓	✓
Data scaling						
Input scaling	Standard	✓	✓	✓	✓	*
	Quantile $\mathcal{U}$		✓	✓		*
	Quantile $\mathcal{N}$		✓	✓		*
Output scaling	Standard	✓	✓	✓	✓	*
	Quantile $\mathcal{U}$		✓	✓		*
	Quantile $\mathcal{N}$		✓	✓		*
Sampling and data generation						
Number of classes	{2, 3, 4}			✓	**	**
Purposeful sampling	{No, 2, 3, 4}	No	✓	✓	✓	*
Upsampling	{Yes, No}	No	✓	✓	✓	*
Specific to the mixture-of-experts						
Weighted opinions	{Yes, No}					✓
Hyperparameter Optimization algorithm						
HPO	{Exhaustive, Bayesian}	Ex	B	B	B	B

Table 6

Hyperparameters and ranges for the tree boosting models.

Name	Options/range	Tree boosting	
		GBDT	XGB
<i>GBDT / XGB hyperparameters</i>			
Nb Estimators	[64, 512]	✓	✓
Learning rate	$\log(lr) \in [-5, -1]$	✓	✓
Max Tree Depth	[3, 8]	✓	✓
Subsample frac	[0, 1]	✓	✓
Nb Features	[2, 5]	✓	✓
Min Child Weight	[1, 10]	✓	✓
Min Loss Reduction	[0, 5]	✓	✓
Weights balance	[1, 10]		✓
L1 Regularization	$\log(\alpha) \in [-5, -1]$		✓
L2 Regularization	$\log(\lambda) \in [-5, -1]$		✓
Huber Regularization	[0, 1]	✓	
<i>Data scaling</i>			
Input scaling	Standard	✓	✓
	Quantile $\mathcal{U}$	✓	✓
	Quantile $\mathcal{N}$	✓	✓
Output scaling	Standard	✓	✓
	Quantile $\mathcal{U}$	✓	✓
	Quantile $\mathcal{N}$	✓	✓
<i>Sampling and data generation</i>			
Number of classes	{2, 3, 4}		
Purposeful sampling	{No, 2, 3, 4}	✓	✓
Upsampling	{Yes, No}	✓	✓
<i>Hyperparameter Optimization algorithm</i>			
HPO	{Exhaustive, Bayesian}	B	B

3.4. Evaluation

3.4.1. Metrics

The evaluation of classifiers uses the accuracy metric, i.e., The ratio of correctly classified samples in the set. The evaluation of regressions uses two unitless metrics: a standardized mean absolute error (SMAE)

³ <https://scikit-optimize.github.io>.

⁴ <https://scikit-learn.org/>.

⁵ The classifier used for sampling (Fig. 4) also undergoes an HPO, similar to these of the Base model, with only the *small* architecture.

(Eq. 11) and the coefficient of determination (R^2) (Eq. 12), used as reference in all related literature [26–29]. The error metrics involve the predicted value $\hat{y}_i$, the true value y_i and the average of the true values in a dataset $\bar{y}$.

$$SMAE = \frac{\frac{1}{n} \sum_i |y_i - \hat{y}_i|}{\max(y) - \min(y)} \quad (11)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (12)$$

The SMAE is standardized with the range of target values (Eq. 11). The standardization was not done with the average, as the average of some strongly imbalanced datasets approaches zero, magnifying the error. The MAPE (percentage error for each sample) is discarded, as most cases contain samples with zero values. The SMAE as described in Eq. (11) appeared as the best metric to ensure comparability between all cases (c.f. list in Section 2.3). Moreover, the SMAE is easily interpretable, and can be simply converted into the MAE metric, which is interpretable in the physical unit of the predicted variable.

3.4.2. Evaluation procedure

Each model is trained and evaluated four times at each iteration of the HPO and during the final evaluation. This aims to limit the variability induced by the random initializations [26].

During the HPO, the robust performance measure sums the average and standard deviation of the SMAE for regressions (Eq. 13) and of the accuracy for classifications (Eq. 14) on the validation set.

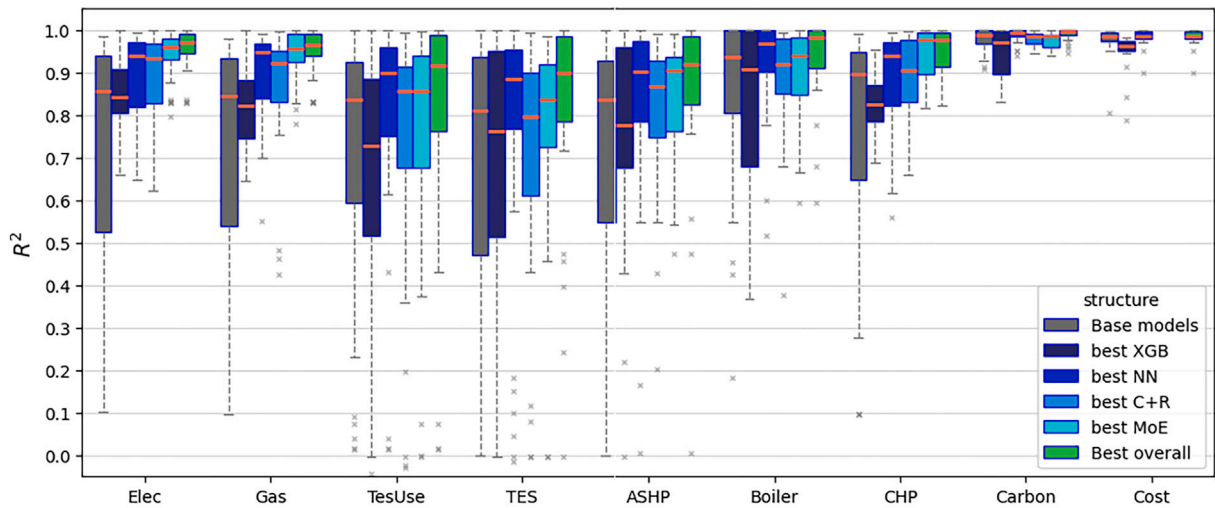
The final reported performance is the average value of each metric on the predictions of all four models on the test set. Additionally, the prediction of the ensemble (c.f. Section 3.2.5) is evaluated similarly.

$$score_{HPO,reg} = average(SMAE) + std(SMAE) \quad (13)$$

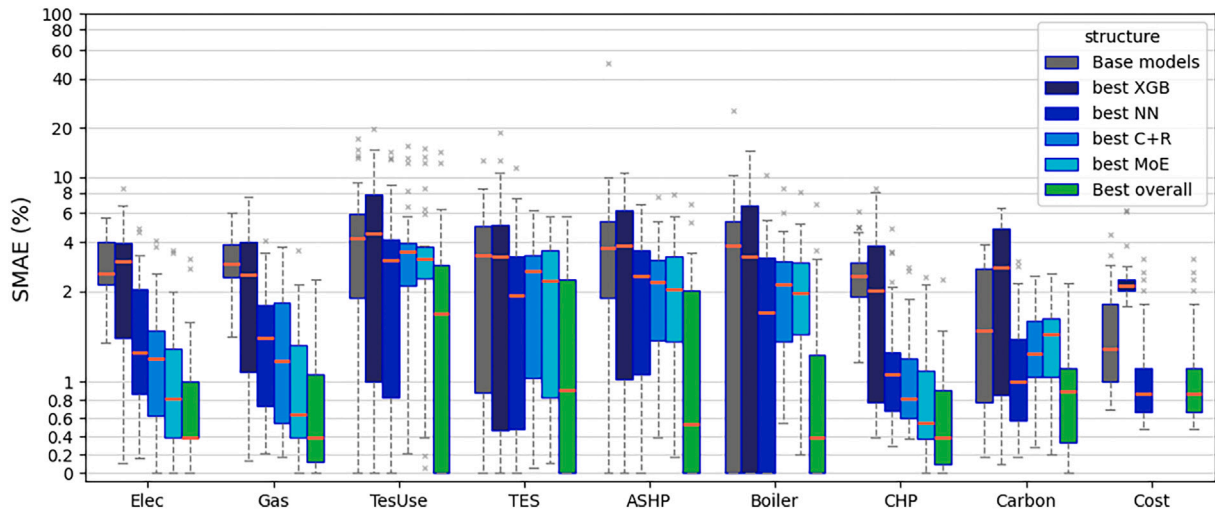
$$score_{HPO,clf} = average(Accuracy) + std(Accuracy) \quad (14)$$

4. Results

The benefits of the novel approach are presented in three parts. Section 4.1 covers the performance across all cases. Section 4.2 illustrates the effect of the approach via specific examples. Section 4.3 validates the approach with regard to the initial hypothesis.



(a) Coefficient of determination R^2



(b) Standardized Mean Absolute Error SMAE

Fig. 7. Performance improvement of each architecture and with the improved methodology.

4.1. Does it work?

Fig. 7(a) and (b) compares the performance of the base model, tree boosting methods (unified under “XGB”), each of the three NN-based architectures and the overall best improved model for all considered cases and variables. Note that only the NN and XGB structures were considered for the cost during the improvement process, due to already strong performance of the base model.

Overall, surrogates of all three NN-based structures consistently outperform base models, with only a few cases showing no improvement. Selecting the best among the three NN-based model types for each case shows significant performance improvements over each individual model type across the pool of cases. There is no single best architecture across all cases. However the XGB models consistently give equivalent or worse results than the base models.

The R^2 metric highlights the strong performance of base models on Cost and Carbon (median ≥ 0.99). It also shows lower medians (between 0.8 and 0.93), with lower quartiles around 0.5 for other variables. This confirms the large variability of performance observed in related literature, since in 25 % of cases, the model is only able to explain 50 % of the variance in the data, leaving 50 % unexplained. The improved approach significantly increases performance; for the best among all model types, medians rise above 0.9 for ASHP and storage-related variables, above 0.96 for others, and the 75th percentiles rise above a R^2 of 0.75 (Fig. 7(a)). The MoE seems to perform overall better than other model types for Elec, Gas and CHP, and the NN better for the Boiler, Carbon,

Cost and storage related variables. However, the best among three model types is significantly better than each individual type.

The SMAE further increases the impression of clear superiority of the best-among three over each individual model type. The base models show a median between 2 % and 4 % SMAE for most variables. It slightly exceeds 4 % for TesUse and lies between 1 % and 2 % for Cost and Carbon. The improved process drops the median SMAE below 1 % for all variables except TesUse (1.7 %), with the 75th percentile only exceeding 2 % SMAE for the ASHP and storage-related variables (Fig. 7(b)).

Despite clear improvements with the novel approach, performance variability remains, particularly for the ASHP capacity and storage-related variables. Moreover, the metrics show similar trends but differ greatly in intensity. For example, the performance on Cost and Carbon stands out with the R^2 metric, while they do not among improved models with the SMAE.

Fig. 8 shows the frequency of usage of data preprocessing, alternative sampling and model structures among the best performing improved models.

Standard scaling (ϵ) is largely preferred to uniform (U) and normal (N) scaling through quantile transform for the inputs and for Cost and Carbon target variables. Frequent usage of all three approaches for other target variables suggest that the choice of scaling depends on each specific dataset.

Purposeful sampling is used for all variables, the least frequent being for the Cost. Upsampling is also frequently used, although less than

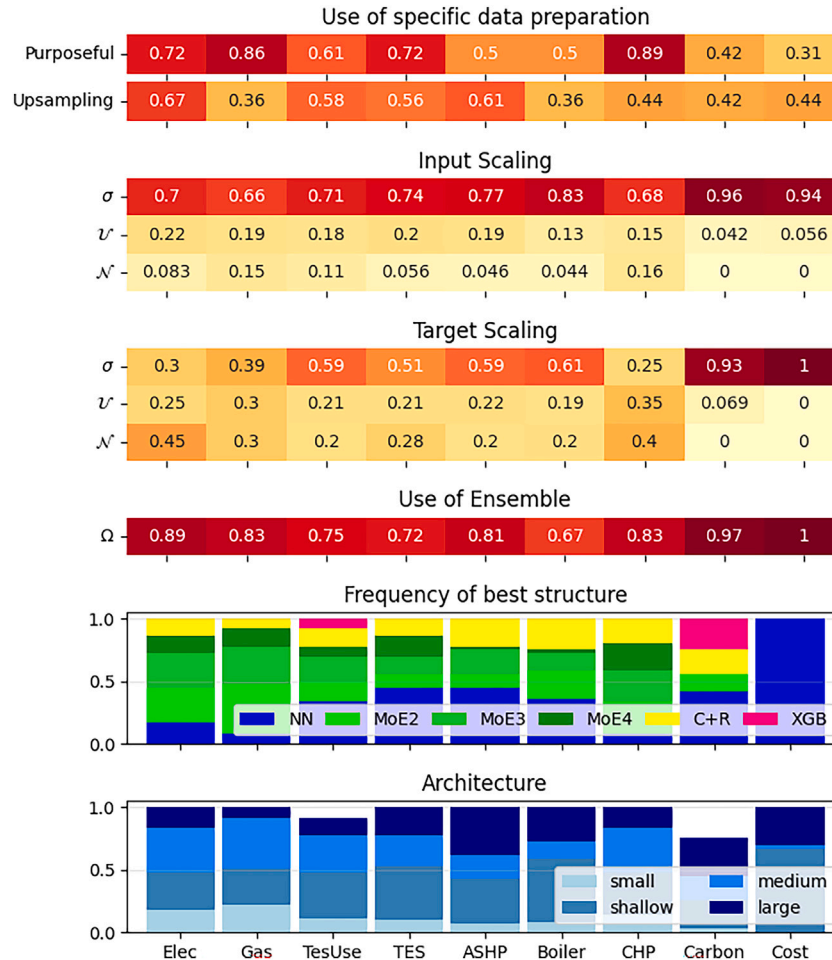


Fig. 8. Frequency of usage of data sampling, processing and model types among the best performing improved models.

purposeful sampling. Both appear as mostly beneficial, however without guaranteeing optimal performance for all datasets.

The model preference is highly variable specific. MoE is preferred for energy purchases and CHP capacity, while the fine NN is preferred for the already well-predicted Cost and Carbon variables. Both models are used with comparable frequencies for the ASHP, Boiler and storage variables. The C + R is never preferred for the Cost and Gas, and is otherwise used in 10 %–20 % of cases, but never as the predominant model.

The XGB is only preferred for 25 % of the carbon emissions predictions (9 cases), all for a relative decarbonization goal of -100% (i.e., at minimum CO_2 emissions). The corresponding base models already yield $R^2 \geq 0.999$. There, emissions are sensitive to only one variable (the carbon intensity of electricity). Improvements are marginal and for a reduced set of particular cases. Therefore, the XGB is discarded for the rest of the analysis. Appendix A further discusses the inadequacy of tree-based models to the present experiment.

Fig. 9 quantifies how superior each model type is in each case, detailed per model type and variable. Where NN is the overall best (Fig. 9(a)), the difference with MoE and C + R remains low ($< 0.5\%$ SMAE median, 25th percentile close to zero). When C + R is the overall best (Fig. 9(b)) the median difference with MoE is below 0.25% SMAE, larger variability and a median difference up to 0.75% SMAE is found with the NN. When MoE performs best (Fig. 9(c)), the median difference with the best NN exceeds 0.5% SMAE, with 75th percentiles consistently between 1% and 1.5% SMAE. The difference is less with the best C + R

but remains significant. Overall, the MoE structure appears as the most reliable, showing significant benefits over peers when performing better (Fig. 9(c)) and a moderate disadvantage otherwise (Fig. 9(a) and (b)).

Overall, all of the suggested architectures and methodological improvements significantly outperform the base case on capacity variables needed for system design. Selecting the best-of-three improved models gives the highest performance. If only one architecture is to be chosen, the Mixture-of-Experts (MoE) model performs the best, achieving median results within 0.5% SMAE of the other architectures.

4.2. How does it work?

The overall experiment evaluates the base model, three architectures and the best overall models through 324 cases. Figs. 10–14 show predictions on five cases where the novel methodology, and particularly MoEs, shows clear improvements. Each figure compares true and predicted values. The left-hand graph shows the sorted true values and their corresponding predictions. The right-hand graph shows the true versus predicted values. Colored halos indicate the MoE-based clustering; samples misclassified by the MoE are marked and corrected in red. Performance measurements are included in a table, including the average relative error induced by misclassified samples (ErrAvg %) and performance if all samples were correctly classified (Perfect).

For variables exhibiting continuous distributions (Fig. 10), like the Cost and Carbon, the base NN already provides accurate predictions.

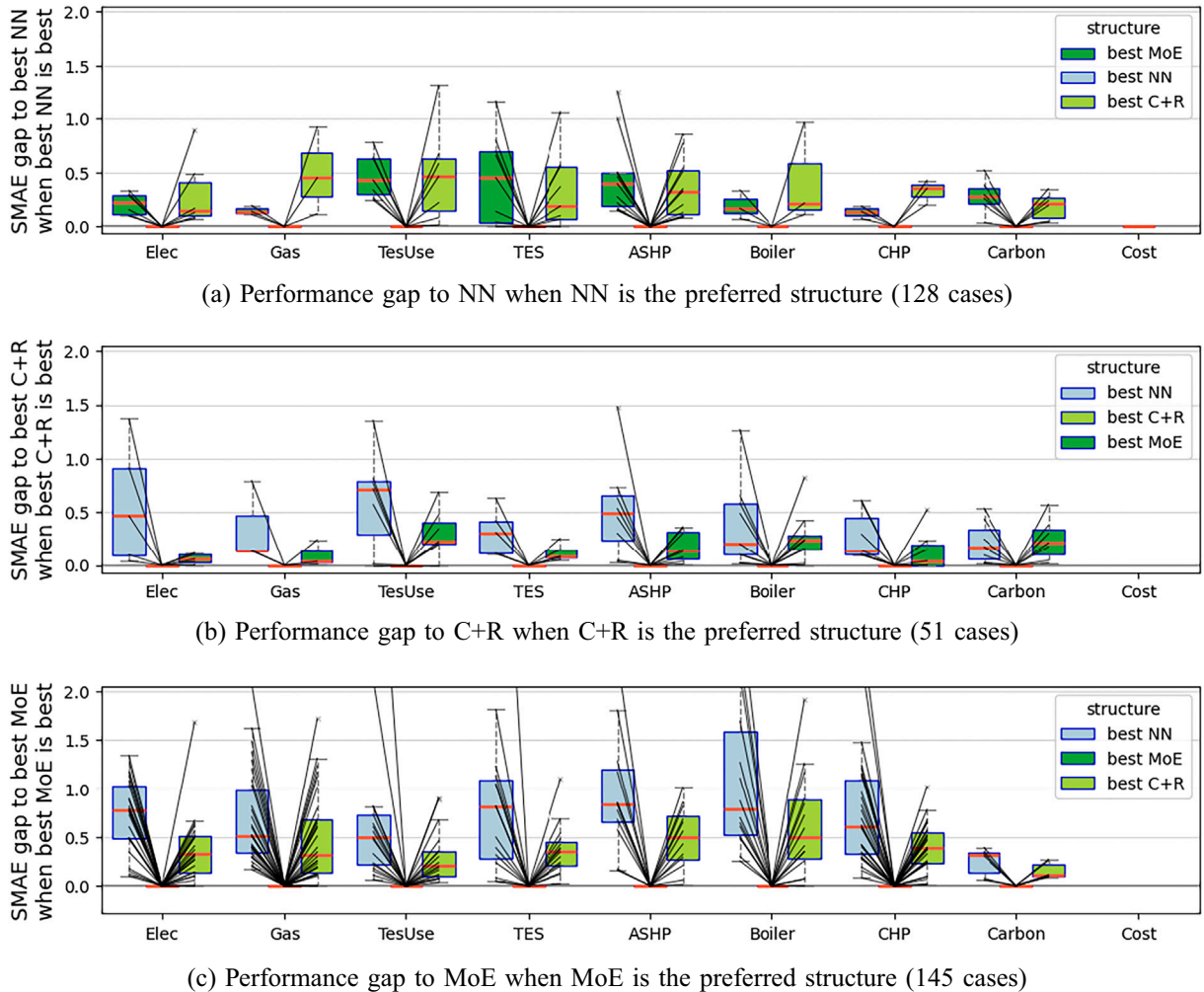


Fig. 9. Difference in performance between the overall best and the best of each architecture.

In this example, the MoE structure slightly outperforms the base NN. It further halves the already low SMAE, despite misclassifications with minor consequences due to the continuity of the distribution.

Fig. 11 shows a case exhibiting a clear discontinuity, which is well captured by the MoE structure. The improved model mainly addresses the errors of the base NN near the transition between the two data segments. Implausible negative predictions from the base NN are also addressed.

The case in Fig. 12 shows a clear discontinuity with strong imbalance between classes. An MoE, via the improved methodology, perfectly predicts the variable whereas the base NN shows errors similar to the case in Fig. 11. Through these two cases, the R^2 metric is inconsistent; two drastically different scores are obtained for the base NN despite similar behaviors and SMAEs in both cases.

Fig. 13 combines both a split and some marginal behaviors. An MoE is best suited, halving the SMAE achieved by the base NN. The MoE dedicates two experts to capture marginal behaviors, and two more to distinguish the two main segments. Misclassifications remain between the two main segments, but their relative proximity limits their influence on the performance.

Fig. 14 shows two major splits in the data distribution, for which the base NN handles the transitions poorly. The MoE halves the SMAE,

however few misclassified samples strongly impact the performance. While the structure of the data suggests using 3 experts, a model with 2 experts was preferred, performing better particularly due to a better classification. Fig. 14 also shows how R^2 can mask poor predictions on split data distributions, as 2 % misclassification on the MoE is valued equally to a base NN capturing the data structure incorrectly.

These examples illustrate the superiority of MoE structures and the improved methodology over the base NN, particularly in handling splits and marginal behaviors. The MoE's ability to segment the data and suggest split predictions addresses the tendency of base NNs to smooth out the transitions between classes.

The examples also suggest that MoEs are limited by misclassifications, as each sample may yield large errors strongly impacting performance. Fig. 15(a) confirms this trend globally, revealing a correlation with the SMAE. Interestingly, identical correlation slopes reveal slightly better MoE performance with more experts, while simultaneously reducing the strength of the correlation slightly.

These examples also illustrate how the R^2 metric often inadequately reflects model performance on data exhibiting splits. The absence of trends between the R^2 performance and misclassifications observed in Fig. 15(b), while clear trends are obtained with the SMAE, confirm this.

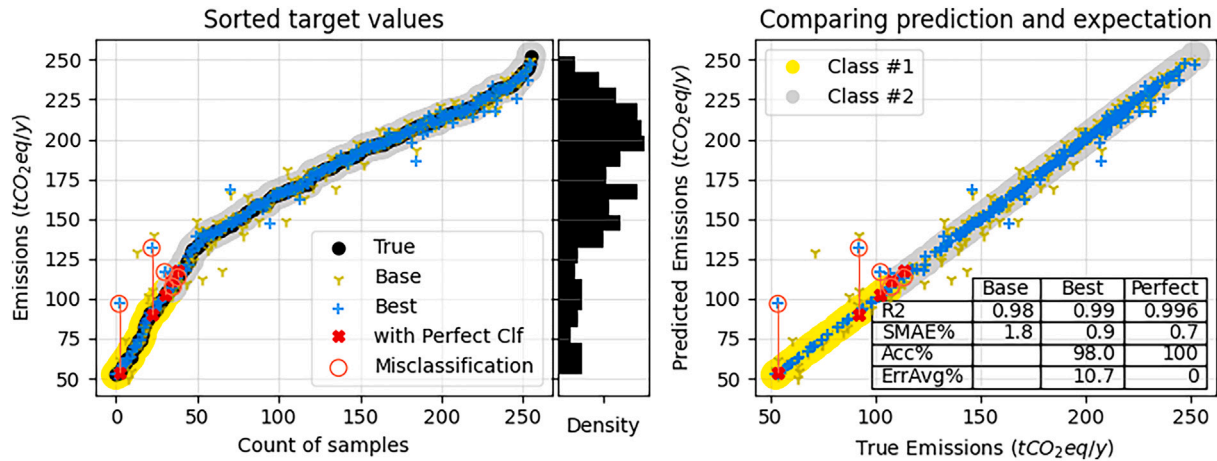


Fig. 10. Carbon emissions output for a MediumOffice in San Francisco at $-50\%CO_2$.

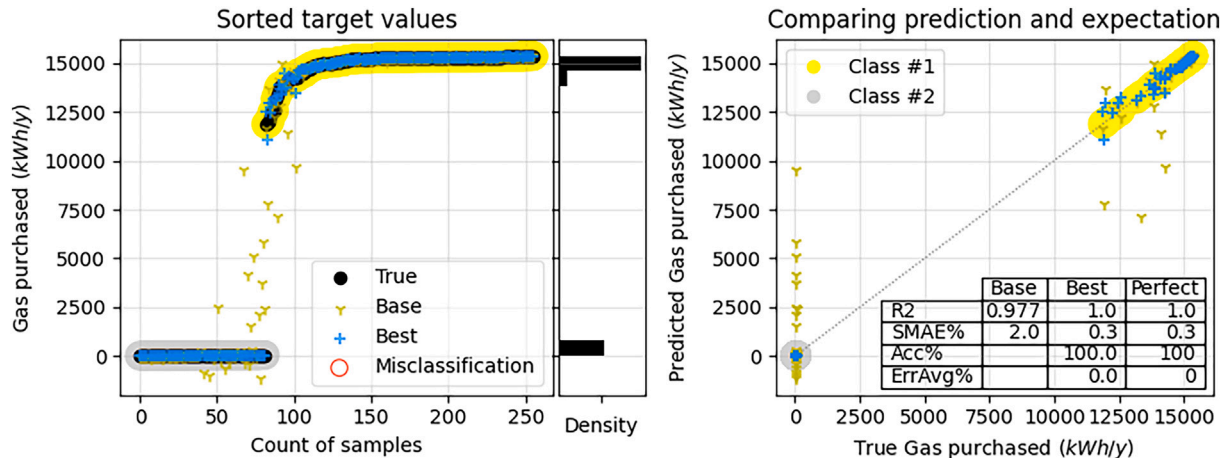


Fig. 11. Gas output for a Hospital in Miami at minimum cost.

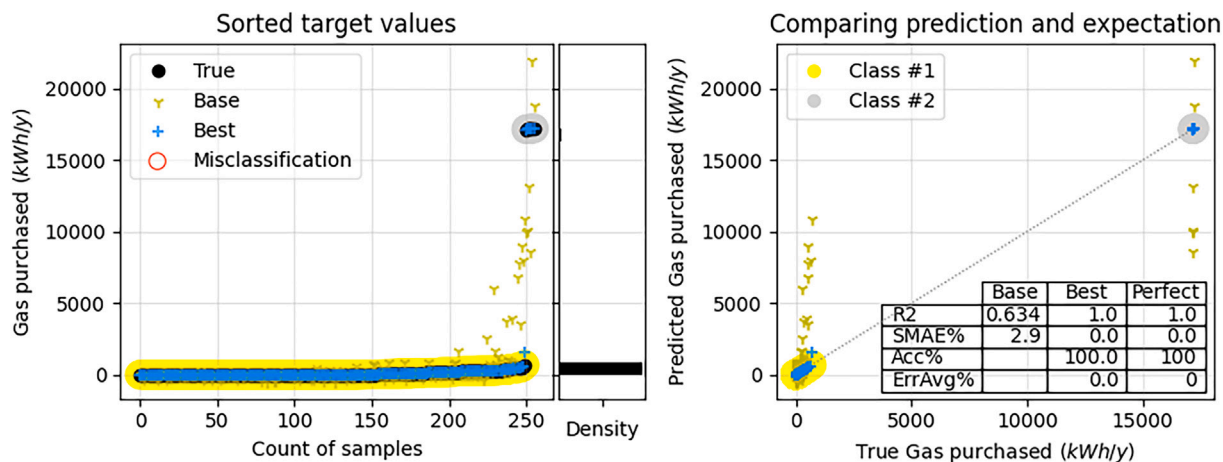


Fig. 12. Gas output for a hospital in Helena at minimum emissions.

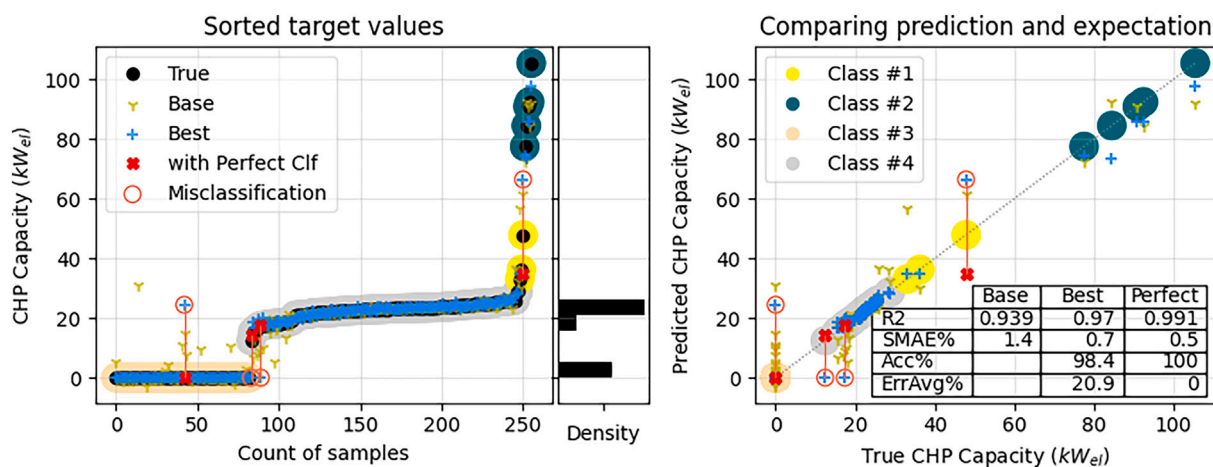
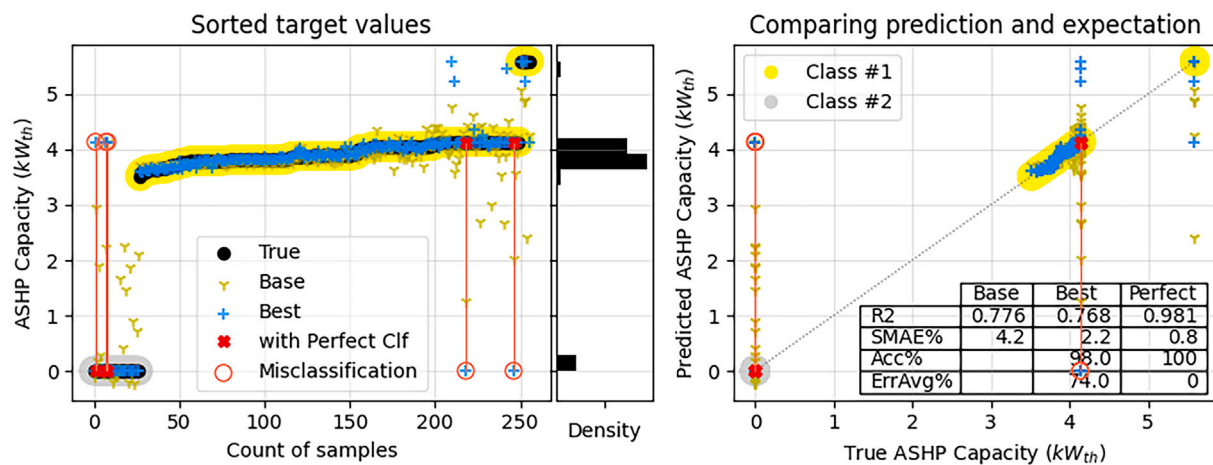
Fig. 13. CHP capacity for a Medium Office in San Francisco at $-50\% \text{CO}_2$.

Fig. 14. ASHP capacity for a single-family house in San Francisco at minimum cost.

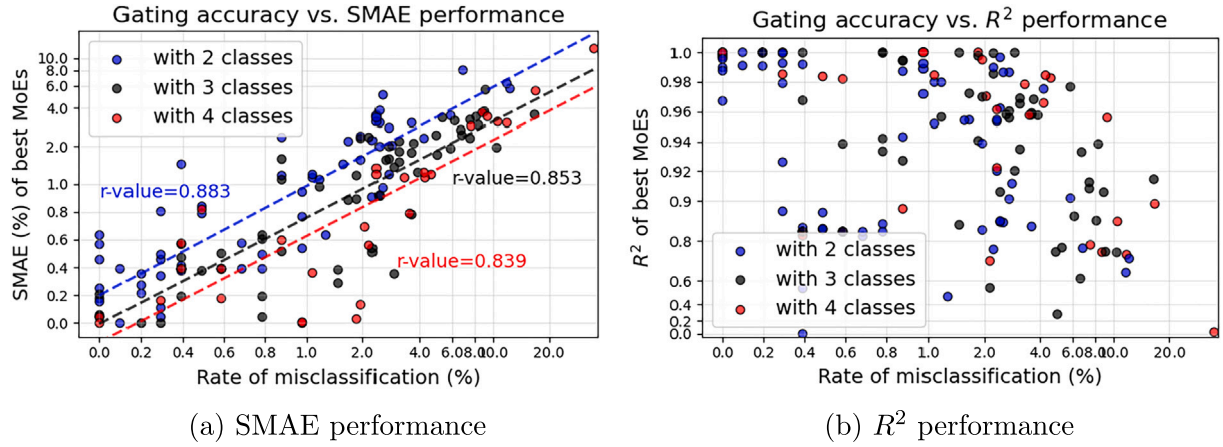


Fig. 15. Link between global performance and misclassification.

4.3. Why does it work?

The hypothesis behind this work suggests that ANN surrogates may perform better with normally or smoothly distributed target values. Fig. 16 displays results of a clustering process grouping cases by similarity of distribution. The clustering is applied to the sorted target values of base cases, and then improved cases are assigned to these clusters after being transformed as during the training phase. The full clustering procedure is described in Appendix B.

The left column of Fig. 16 shows the distribution of SMAE achieved by the base and the best improved models on each case of a cluster. Overlaying points show the SMAE achieved for each, distinguishing between base and improved models, sorted vertically by decreasing performance, and aligned horizontally on the SMAE.

Nine clusters were identified. Cluster 0 stands out, by capturing one-third of all base cases but two-thirds of the improved cases. It shows a smooth distribution of values, with a peak of density between 0 % and 1 % SMAE, and rapid decline as the error grows. It shows that the novel approach pushes cases towards this cluster, increasing the chances of good performance.

Clusters 2, 3 and 4 represent cases with split distributions. Two-thirds of the base cases are reassigned to other clusters. These distributions are challenging for surrogates, as the SMAE spreads densely up to 5 % in clusters 3 and 4, and peaks at 3 % in cluster 2. The remaining improved cases show the best performance in their clusters.

Clusters 5 and 6 represent cases displaying marginal behaviors, and show SMAEs spreading densely up to 6 %. Of the 68 base cases, the 20 remaining after improvement are mostly MoE models able to capture marginal behaviors adequately and perform well.

Cluster 1 represents cases with continuous yet heavily skewed distributions and a large spread of error. 50 % fewer cases are assigned to Cluster 1 after improvement, most of them with an SMAE below 1 %, although some remain with errors within 3 %–5 %. This confirms that the heavy skewness makes cases challenging but that the novel approach may correct it.

Table 7 shows the distribution of base and improved cases across clusters for each target variable. Cost and Carbon are entirely in Cluster 0 from the base cases, explaining their strong initial performance. For all other variables, the novel approach increases the number of cases assigned to Cluster 0. This supports both the hypothesis of this research and the increase in performance observed in Section 4.1.

For base cases, most variables are concentrated in specific clusters: ASHP and TesUse in Cluster 1, Boiler in Clusters 0 and 7, and CHP in Cluster 6. TES spreads across Clusters 0, 1, 6, and 7. This shows that most variables have their own specific distribution.

Interestingly, at least a third of improved cases assigned to Cluster 1 are newly assigned, coming from clusters with marginal behaviors (Cluster 5 for Elec, Cluster 6 for CHP). This indicates that the novel approach may result in heavily skewed distributions for cases containing a few marginal behaviors.

Table 7
Count of cluster assignments per variable.

Method	Cluster																	
	0		1		2		3		4		5		6		7		8	
	Base	Improved	Base	Improved	Base	Improved	Base	Improved	Base	Improved	Base	Improved	Base	Improved	Base	Improved	Base	Improved
Elec	4	25		2		12	3	2		4	17	1						2
Gas	1	25	2	2			1		14	2		2	18	5				
TesUse	12	24	8	1	1		1	1	1	2	2		2	1	5	6	4	1
TES	10	20	8			1			3	2			7	6	7	7	1	
ASHP	16	25	6	2	1		1	1	1		1		3	2	5	5	2	1
Boiler	16	21	2	2		2	3		3						10	10	2	1
CHP	3	24	3	6			2	2	6	1			18	3			4	
Carbon	36	36																
Cost	36	36																

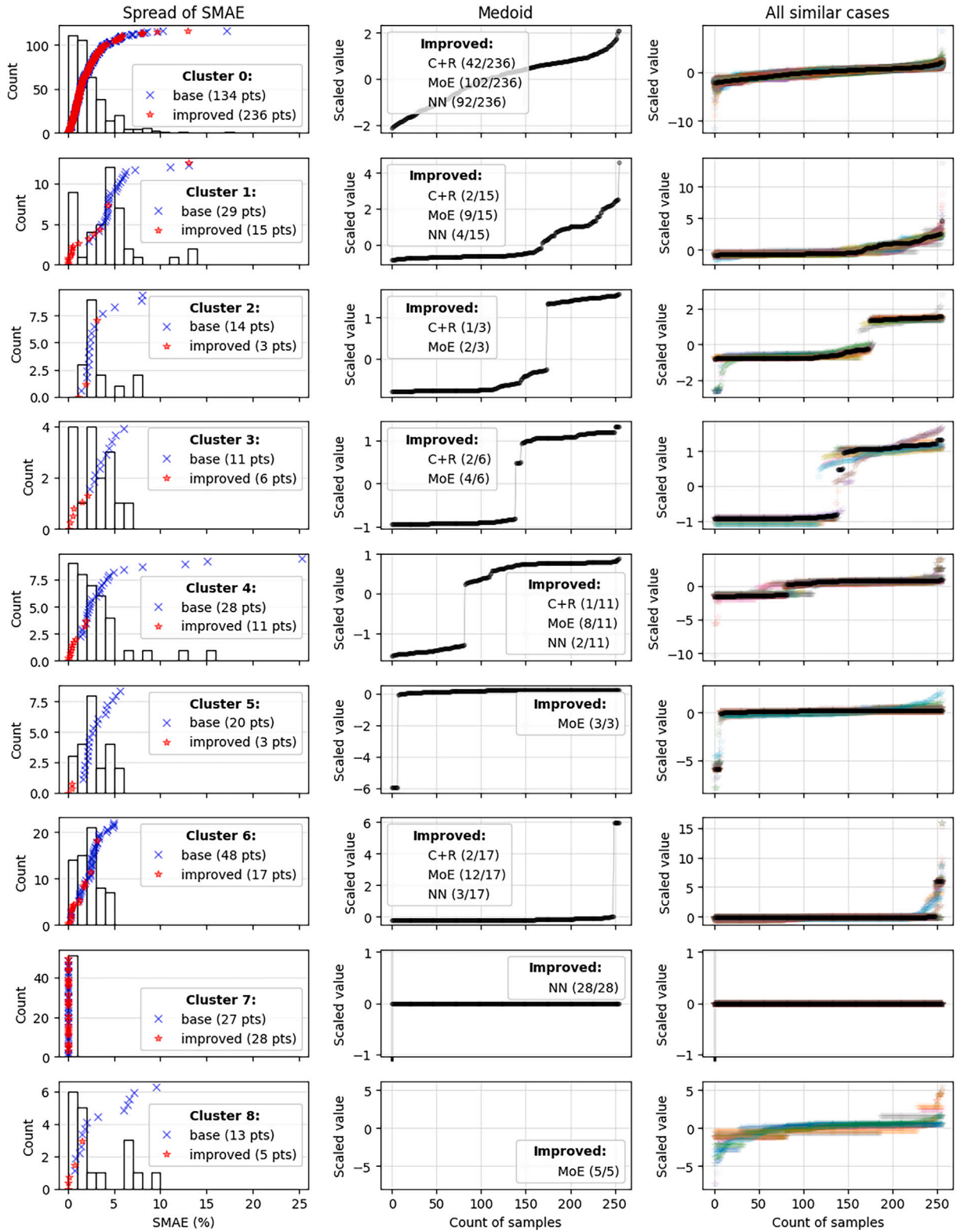


Fig. 16. Sorted values of design variables and scores of all cases, clustered by the distribution of the scaled output value across the cases. The left column shows histograms of performance for assigned cases. The cluster medoids (i.e., most typical case of the cluster) are shown separately in the middle column and with all cases in the same cluster in the right column.

5. Conclusions

This study proposes a methodology to develop surrogate models for the design of multi-energy systems (MES). The methodology combines different data preprocessing, sampling and model structures. These novel methodological components are tailored to address specific properties of the MES design, i.e., design variables often exhibiting splits, skewness or extremely imbalanced distributions. These arise from the underlying physical problem: minor variations of the input may cause abrupt operational changes engendered by a sudden selection, exclusion, or combination of specific technologies.

The improved methodology is evaluated across 324 cases involving diverse design variables and compared to the standard approach from literature. Results show that the improved methodology consistently outperforms the standard approach, particularly in cases with challenging data distributions. Supported by a specific sampling, upsampling, data scaling and architecture, the improved methodology transforms the data distribution perceived by learning components into smoother and better balanced distributions. The best choice of surrogate model architecture is case dependent, but a Mixture-of-Experts (MoE) is most reliable overall. Particularly, the MoE successfully identifies data splits and marginal behaviors, where the standard architecture struggles. This is due to its particular architecture: a gating unit subdividing the datasets, possibly based on irregularities, allowing expert units to learn the smoother data distribution of each segment separately.

A key limitation of the improved methodology lies in the possible misclassifications from the gating unit of MoE structures, which may strongly impact the overall reported performance. The experiment also highlights the inadequacy of the coefficient of determination R^2 to reflect model performance on data exhibiting split and imbalanced distributions.

Future efforts may improve this methodology and extend the application of surrogate modeling for MES to broader contexts. First, the gating of MoEs needs more attention to ensure the highest reliability and robustness of surrogates. Other ensemble methods could be investigated, including expanding the MoE approach to encompass the larger neural network and classifier-regressor architectures used here that sometimes outperform the current MoE. Next, future efforts should focus on expanding the application of each surrogate to more variables and contexts (e.g., across multiple building types, environmental and policy conditions). Similarly, the MoE and other ensemble methods could be applied to other areas of surrogate modeling that exhibit similar issues with split or imbalanced data distributions. Finally, investigating the interconnection between surrogate models

could also extend the modeling capabilities to community-level and beyond. Such improvements could play a crucial role in energy planning and decarbonization strategies on a broader scale.

CRediT authorship contribution statement

François Lédée: Writing – review & editing, writing – original draft, visualization, validation, software, methodology, investigation, formal analysis, and conceptualization. **Curran Crawford:** Writing – review & editing and supervision. **Ralph Evins:** Writing – review & editing, supervision, and funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Incompatible use of decision trees

A novel methodological component is the use of Mixture of Experts (MoE). This architecture successfully identifies splits in the design space (c.f. Section 2.4). The MoE divides the data into subregions to simplify the training and prediction process. Decision trees (DT) and tree-based boosting methods work in a similar manner. However, they partition the input space one variable at-a-time. This may be effective for problems in which one feature explains the variability globally or locally. In surrogate modeling however, and particularly for the design of MES, the design variables often correlate with combinations of inputs over the whole design space, making single-variable splits insufficient.

Ensemble and boosting procedures simply aggregate inadequate models. The added complexity and inadequacy of base learners induce instability, yielding insufficient precision on unseen data near the splits.

To illustrate, a reduced experiment with 2 input features (similar to Fig. 2(a)) is conducted. A full set of 512 samples is generated with Halton sampling, the first 64 samples are used for training and the last 64 are used for testing (Fig. A.1).

The dataset displays a split linearly combining the two variables, with no variability on either side of the split. Therefore, the ability of models to classify the dataset is evaluated. Models are evaluated on both

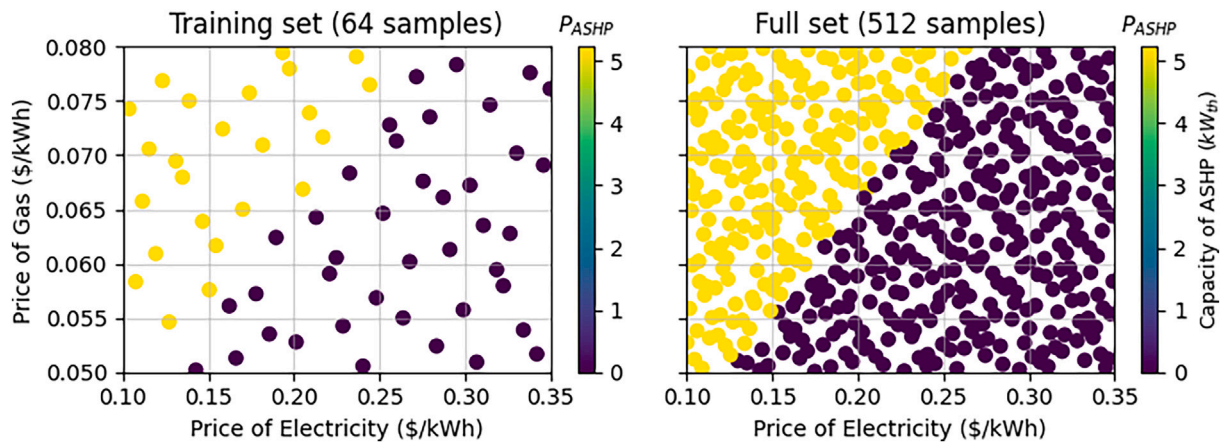
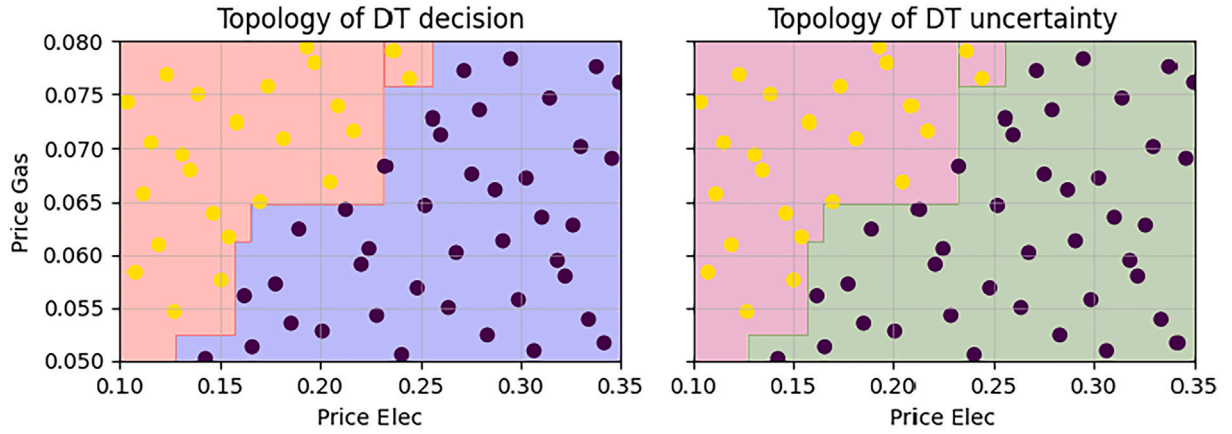
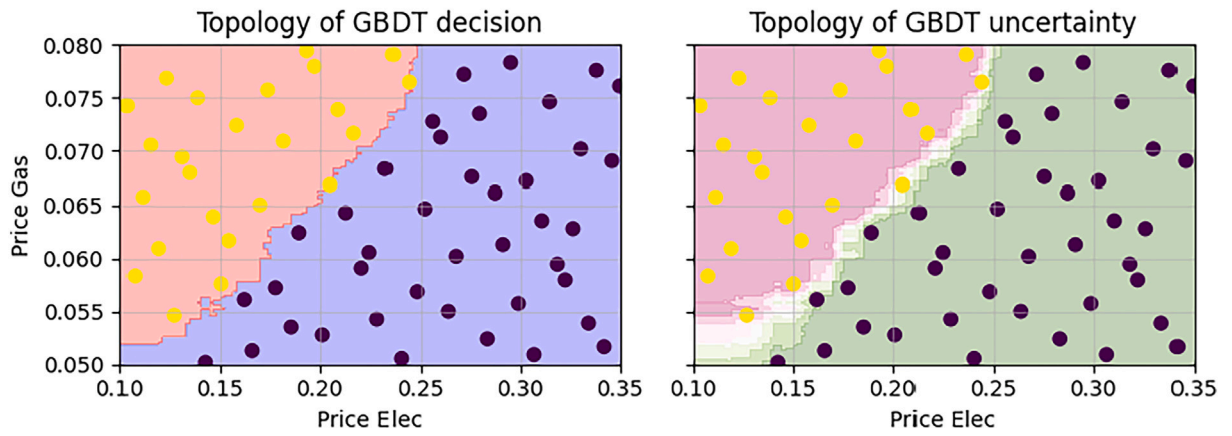


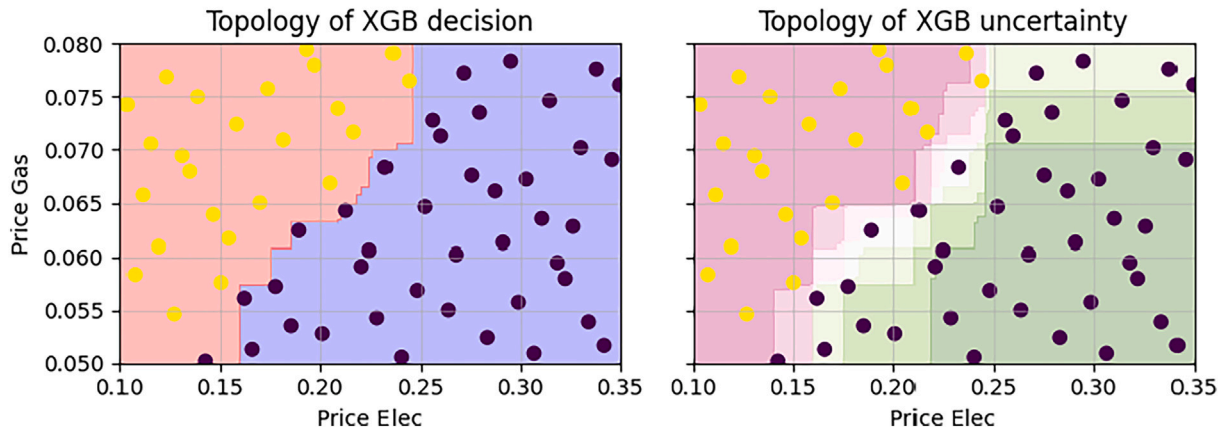
Fig. A.1. Dataset for the reduced 2D experiment.



(a) Decision Tree (DT)



(b) Gradient Boosting Decision Trees (GBDT)



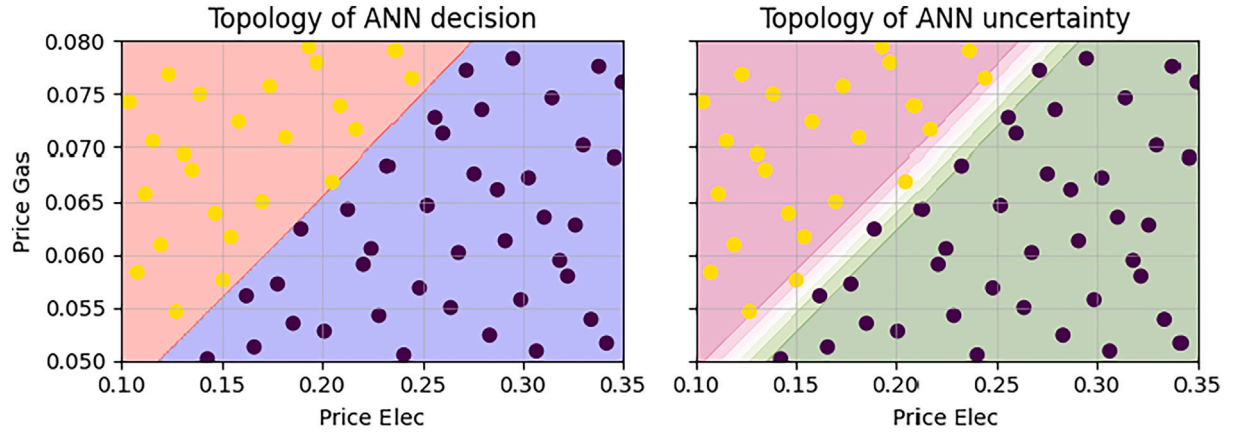
(c) eXtreme Gradient Boosting (XGB)

Fig. A.2. Prediction of models based on decision trees.

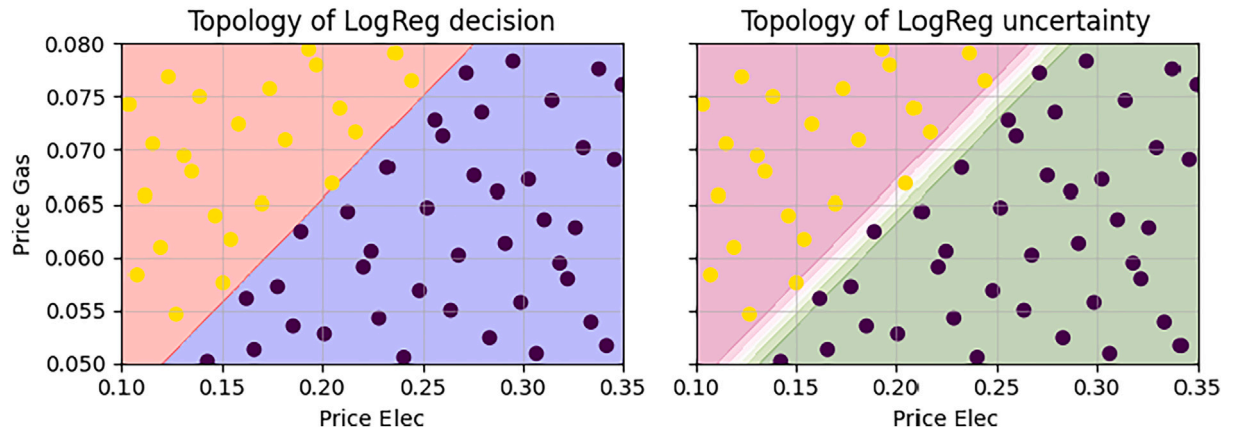
a test set and the full set (highest definition of the split), using the R^2 and $SMAE$ metrics. Note that tree-based models undergo a hyperparameter tuning, others use default hyperparameter values.

The single DT forms a stairs-like decision boundary, exploiting the variations in the density of samples along the split to modulate the

size of steps (Fig. A.2(a)). The effect of ensemble in the GBDT allows for much smaller steps in the stairs-like boundary, as well as some uncertainty in subregions with few samples (Fig. A.2(b)). The resulting boundary is overfitted, and misses the actual split. For the XGB, the regularization mechanisms simplify the boundary in comparison to the



(a) Artificial Neural Network (ANN)



(b) Logistic Regression (LogReg)

Fig. A.3. Prediction of models not based on decision trees.

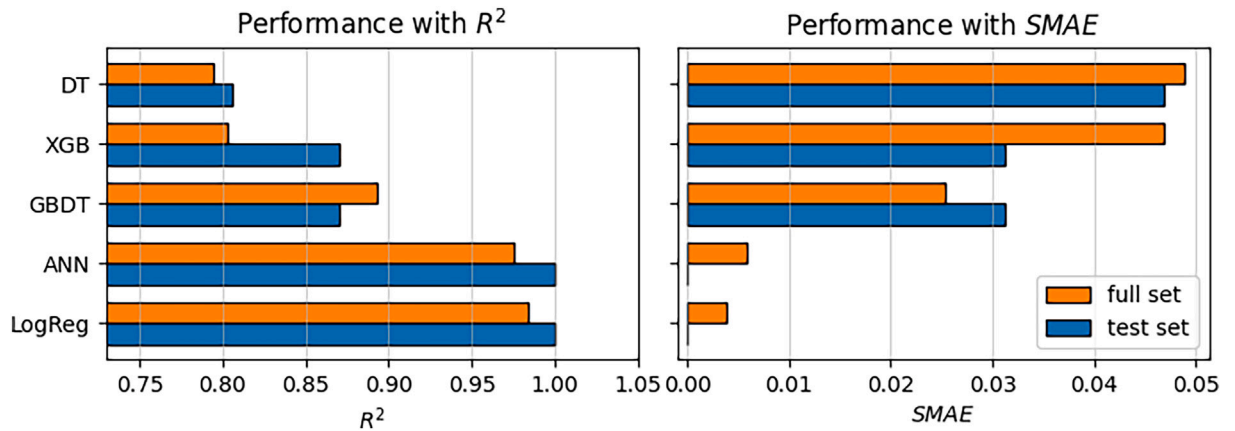


Fig. A.4. Model evaluations for the reduced experiment.

GBM (Fig. A.2(c)). The XGB still draws a stairs-like boundary, with questionable uncertainty patterns and uses least dense subregions to reduce its complexity.

ANNs (Fig. A.3(a)) and LogReg (Fig. A.3(b)) are better adapted to the topology of the MES design problems. The decision boundaries and

uncertainties obtained from the training set align with both the test set and expectations for the full set (Fig. A.1).

This difference in properties highlights why tree-based methods are inadequate in comparison to other models (e.g., ANNs, MoE). Performance evaluations in Fig. A.4 confirm the observations.

Appendix B. Clustering methodology

This appendix explains the clustering algorithm used in Section 4.2 to group cases by similarity. The algorithm consists of three steps: data preprocessing, clustering, and reassignment of outlier samples.

The data preprocessing (Fig. B.1) applies the preferred preprocessing steps for each case to the training target values. The base cases only undergo standard scaling, while the refined cases may involve purposeful sampling, upsampling, a segmentation of the data and specific rescalings. The processed target values are then sorted in increasing order, and an interpolation using a random forest regressor may be employed to reduce the number of samples to 256 if upsampling was applied.

The clustering procedure (Fig. B.2), inspired by [70], applies a k-medoids clustering on the preprocessed data, where similarity is measured using the dynamic time warping (DTW) distance. A case i whose average distance $\bar{d}_{i,c}$ with the rest of its cluster c exceeds a threshold τ_1 are flagged as outliers. The ratio of outliers is compared to critical values (ρ_{min} and ρ_{max}) to decide whether to reassign outliers or the whole cluster to an “exception” cluster, or to carry on a sub-clustering within the cluster. The formation of clusters and decisions with the ratio of outliers only involve base cases, then the refined cases are assigned to corresponding clusters.

Algorithm B.1: Clustering adjustment algorithm.

```

for  $i \in c_{exception}$  do
    Calculate  $\bar{d}_{i,c} \forall c \in C$ ;
    if  $\exists c / \bar{d}_{i,c} < \tau_1$  then
        Assign  $i$  to cluster  $c$  with smallest  $\bar{d}_{i,c}$ 
    else if  $\exists c / (\tau_1 < \bar{d}_{i,c}) \& (\exists j \in c / \bar{d}_{i,j} < \tau_1)$  then
        Assign  $i$  to cluster  $c$  complying with the condition with
        smallest  $\bar{d}_{i,c}$ 
    else
        Leave sample  $i$  in cluster  $c_{exception}$ 
    end
end

```

Finally, samples in the exception cluster are reassessed following Algorithm B.1. This addresses the fact that the clustering step only divides the pool of cases. If the average distance of an outlier to any other cluster $\bar{d}_{i,c}$, or its distance to any other case in another cluster meets the distance threshold $d_{i,c}$, it is reassigned. This reassignment step is completed simultaneously for all cases of the exception cluster to prevent interference among reassigned cases.

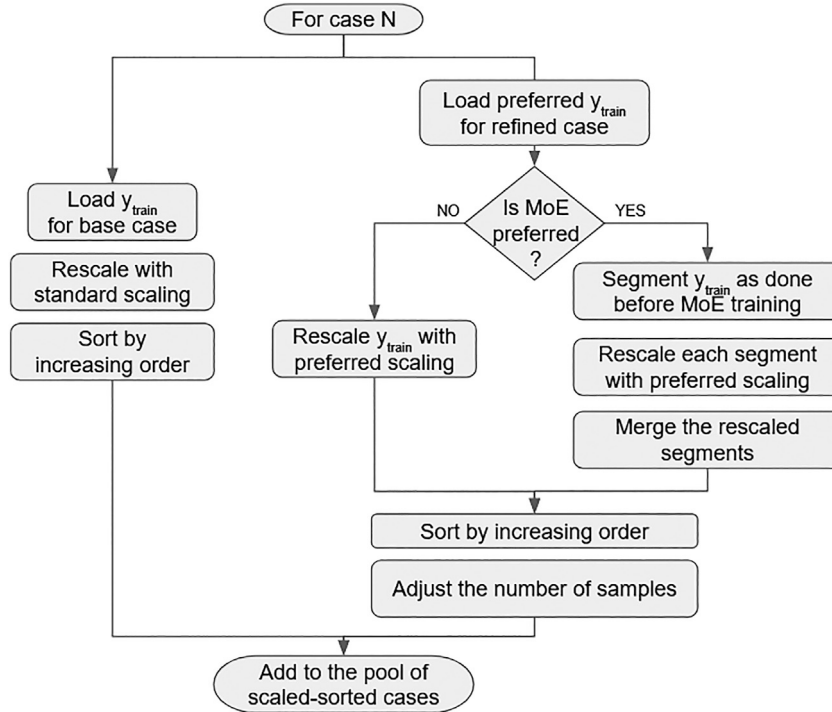


Fig. B.1. Data preparation before clustering to organize cases by similarity.

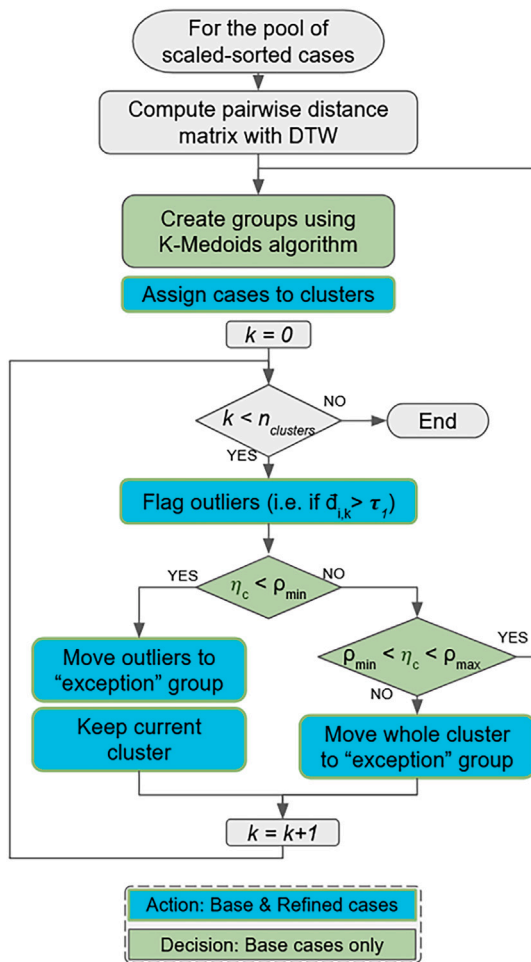


Fig. B.2. Clustering process to group the cases by similarity.

Data availability

The surrogate modeling code is available in the Git repository: <https://gitlab.com/fledee/adaptiveml>. The data and MES algorithms will be made available upon request.

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